

On common $\mathcal{M}$ -subspaces of quadratic (vectorial) Boolean bent functions

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Abstract

In this paper, we investigate the existence of common $\mathcal{M}$ -subspaces among pairs of quadratic Boolean functions and among the component functions of quadratic vectorial bent functions. In particular, we prove that any two quadratic Boolean functions on $\mathbb{F}_2^n$ (including bent functions) share at least $2^{n/2} + 1$ common $\mathcal{M}$ -subspaces of dimension $n/2$. Consequently, we show that every quadratic vectorial $(n, 2)$ -bent function admits, up to EA-equivalence, the canonical form $(x, y) \in \mathbb{F}_2^{\frac{n}{2}} \times \mathbb{F}_2^{\frac{n}{2}} \mapsto (x \cdot y + g(y), x \cdot L(y) + h(y))$, where L is a linear complete permutation of $\mathbb{F}_2^{\frac{n}{2}}$ and $g, h \in \mathcal{B}_{\frac{n}{2}}$ satisfy $\deg(g), \deg(h) \leq 2$. On the other hand, we provide explicit constructions of quadratic vectorial $(n, 3)$ -bent functions whose component functions do not share a common $\mathcal{M}$ -subspace of maximal dimension $n/2$, for all $n \geq 6$. This demonstrates the existence of quadratic vectorial bent functions in the completed generalized Maiorana-McFarland class $\mathcal{GM}^\#$ that lie outside the completed extended Maiorana-McFarland class $\mathcal{EM}^\# \subset \mathcal{GM}^\#$.

Key Words: Boolean function, Vectorial bent function, $\mathcal{M}$ -subspace, Totally isotropic subspace, EA-equivalence, Complete permutation.

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1 Introduction

Let $\mathbb{F}_2^n$ be the vector space of dimension n over $\mathbb{F}_2 = \{0, 1\}$, which can be identified with the finite field $\mathbb{F}_{2^n}$. A mapping $F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is called an (n, m) -function. For $m = 1$, a function $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is called a *Boolean function* in n variables. The set of all Boolean functions in n variables is denoted by $\mathcal{B}_n$. A Boolean function on $\mathbb{F}_2^n$ can be uniquely expressed as a polynomial in $\mathbb{F}_2[x_1, \dots, x_n]/(x_1 + x_1^2, \dots, x_n + x_n^2)$. This representation is called the *algebraic normal form* (ANF, for short), that is, $f(x) = \sum_{v \in \mathbb{F}_2^n} c_v \left(\prod_{i=1}^n x_i^{v_i} \right)$, where $x = (x_1, \dots, x_n) \in \mathbb{F}_2^n$, $c_v \in \mathbb{F}_2$ and $v = (v_1, \dots, v_n) \in \mathbb{F}_2^n$. The *algebraic degree* of a Boolean function $f \in \mathcal{B}_n$, denoted by $\deg(f)$, is the maximum Hamming weight $\text{wt}(v)$ of all $v \in \mathbb{F}_2^n$ such that $c_v \neq 0$. Boolean functions of the algebraic degree at most 1 are called *affine*, and of degree 2 are called *quadratic*. The *Walsh transform* $W_f: \mathbb{F}_2^n \rightarrow \mathbb{Z}$ of a Boolean function $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ at $a \in \mathbb{F}_2^n$ is defined by $W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + a \cdot x}$, where $a \cdot x = a_1x_1 + a_2x_2 + \dots + a_nx_n$ is the dot product on $\mathbb{F}_2^n$. A Boolean function $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, where $n = 2k$, is called *bent* if $|W_f(a)| = 2^{n/2}$, for all $a \in \mathbb{F}_2^n$ [Rot76].

Any vectorial (n, m) -function F (i.e., $m \geq 2$) can be written as $F(x) = (f_1(x), \dots, f_m(x))$, where the Boolean functions f_i on $\mathbb{F}_2^n$ are called *coordinate functions* of F . The algebraic degree of a vectorial function is defined as the maximum algebraic degree of its coordinate functions. For (n, m) -functions, the bent property is introduced with the notion of *component functions* $F_b: \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, which are Boolean functions defined by $F_b(x) = b \cdot F(x)$, for $b \in \mathbb{F}_2^m$. An (n, m) -function is called (n, m) -*bent* (*vectorial bent* for $m \geq 2$), if for all $b \in \mathbb{F}_2^m \setminus \{0_m\}$, the component functions F_b are Boolean bent. Vectorial (n, m) -bent functions exist only for $m \leq n/2$; this result is also known as the Nyberg bound [Nyb91]. Note that the bent property is preserved under the extended-affine equivalence, which is defined as follows. Two (n, m) -functions F and F' are called *extended-affine equivalent* (EA-equivalent), if there exist an affine permutation A_1 of $\mathbb{F}_2^n$, an affine permutation A_2 of $\mathbb{F}_2^m$ and an affine (n, m) -function A_3 such that $A_2 \circ F \circ A_1 + A_3 = F'$. We call *derivative* in the direction $a \in \mathbb{F}_2^n$ of an (n, m) -function F the (n, m) -function $D_a F(x) = F(x + a) + F(x)$. The k -th order derivative of F in the directions $a_1, \dots, a_k \in \mathbb{F}_2^n$ is the (n, m) -function $D_{a_1} D_{a_2} \dots D_{a_k} F(x)$. For extensive references on Boolean and vectorial functions, see [Car21].

One of the most important classes of Boolean bent functions is the Maiorana-McFarland class $\mathcal{M}$, which is the set of n -variable ($n = 2k$) Boolean bent functions of the form $f(x, y) = x \cdot \pi(y) + h(y)$, for all $x, y \in \mathbb{F}_2^{n/2}$, where π is a permutation on $\mathbb{F}_2^{n/2}$, and $h \in \mathcal{B}_{n/2}$ is an arbitrary Boolean function. The set of bent functions $f \in \mathcal{B}_n$ which are EA-equivalent to at least one function in $\mathcal{M}$ is called the *completed Maiorana-McFarland class*

and is denoted by $\mathcal{M}^\#$. With the following criterion of Dillon, one can show that a given Boolean bent function $f \in \mathcal{B}_n$ is (not) a member of $\mathcal{M}^\#$.

Lemma 1.1. [Dil74, p. 102] *Let $n = 2k$. A Boolean bent function $f \in \mathcal{B}_n$ belongs to $\mathcal{M}^\#$ if and only if there exists an $\frac{n}{2}$ -dimensional linear subspace V of $\mathbb{F}_2^n$ such that, for any $a, b \in V$, $D_a D_b f(x) = f(x) + f(x+a) + f(x+b) + f(x+a+b) = 0$, for all $x \in \mathbb{F}_2^n$.*

Following [PP20], we call a linear subspace U of $\mathbb{F}_2^n$ an $\mathcal{M}$ -subspace of $f \in \mathcal{B}_n$, if for all $a, b \in U$ we have that $D_a D_b f = 0$. For a bent function in n variables, the maximum dimension of an $\mathcal{M}$ -subspace is $n/2$; such an $\mathcal{M}$ -subspace of a bent function $f \in \mathcal{B}_n$ will be called *maximal* throughout this article. With this terminology, a bent function $f \in \mathcal{B}_n$ belongs to $\mathcal{M}^\#$ iff f has at least one maximal $\mathcal{M}$ -subspace. For vectorial bent functions, several notions of vectorial Maiorana-McFarland bent functions were considered in [Mes16, Section 12.1.3]. The *Strict Maiorana-McFarland class* $\mathcal{SM}$ is the set of bent functions of the form $F(x, y) = L(x\pi(y)) + G(y)$, where π is a permutation of $\mathbb{F}_2^{n/2}$, L is a linear or affine function from $\mathbb{F}_2^{n/2}$ onto $\mathbb{F}_2^m$, and G is an arbitrary $(n/2, m)$ -function. Recall that an (n, m) -function is called balanced if it takes every value of $\mathbb{F}_2^m$ the same number of times.

The *Extended Maiorana-McFarland class* $\mathcal{EM}$ is the set of bent functions of the form $F(x, y) = \psi(x, y) + G(y)$, where $G: \mathbb{F}_2^{n/2} \rightarrow \mathbb{F}_2^m$ is arbitrary and $\psi: \mathbb{F}_2^{n/2} \times \mathbb{F}_2^{n/2} \rightarrow \mathbb{F}_2^m$ satisfies that, for every y , the mapping $x \mapsto \psi(x, y)$ is linear and, for every nonzero x , the mapping $y \mapsto \psi(x, y)$ is balanced. The *General Maiorana-McFarland class* $\mathcal{GM}$ is the set of (n, m) -functions F such that, for every $b \in \mathbb{F}_2^m \setminus \{0_m\}$, the component function F_b belongs to $\mathcal{M}^\#$.

The completed versions (i.e., closures under EA-equivalence) of these vectorial Maiorana-McFarland classes are denoted by $\mathcal{SM}^\#$, $\mathcal{EM}^\#$, $\mathcal{GM}^\#$, respectively. While several papers dedicated to $\mathcal{M}$ -subspaces have recently appeared in the context of classification and construction of Boolean bent functions [PP20, PPKZ24, KPPZ25, KPP⁺25, KPPZ26, KB26], including the survey article [PPZK26], only a few works [PP21, PZKW21, BPPP24] have considered the vectorial case.

The main aim of this paper is to introduce the theory of $\mathcal{M}$ -subspaces for vectorial (bent) functions, and to investigate in more detail classes of quadratic vectorial bent functions, whose coordinate functions may or may not share common $\mathcal{M}$ -subspaces, thus distinguishing vectorial bent functions in the extended $\mathcal{EM}$ and general $\mathcal{GM}$ Maiorana-McFarland classes.

The rest of the article is organized as follows. In Section 2, we extend several well-known results on $\mathcal{M}$ -subspaces of Boolean functions to the vectorial setting and explicitly describe the elements of the extended Maiorana-McFarland class. In Section 3, we show that the coordinate functions of quadratic vectorial $(n, 3)$ -bent functions do not necessarily share

$\mathcal{M}$ -subspaces of dimension $n/2$. In contrast, we prove in Section 4 that any two quadratic Boolean functions (including bent functions) on $\mathbb{F}_2^n$ share a common $\mathcal{M}$ -subspace of dimension $n/2$ (Theorem 4.2). Moreover, we show that, when n is even, there exist at least $2^{n/2} + 1$ common $\mathcal{M}$ -subspaces of dimension $n/2$ (Theorem 4.4). As a consequence of Theorem 4.2, we show that all quadratic vectorial $(n, 2)$ -bent functions belong to $\mathcal{EM}^\#$ and specify their canonical form (Corollary 4.3). The paper is concluded in Section 5.

2 $\mathcal{M}$ -subspaces of vectorial (bent) functions, and the extended Maiorana-McFarland class

As in the Boolean case, we call a linear subspace U of $\mathbb{F}_2^n$ an $\mathcal{M}$ -subspace of an (n, m) -function F , if for all $a, b \in U$ the second-order derivatives satisfy $D_a D_b F(x) = F(x) + F(x + a) + F(x + b) + F(x + a + b) = 0_m$, for all $x \in \mathbb{F}_2^n$. The following straightforward statement indicates that $\mathcal{M}$ -subspaces of vectorial functions are common $\mathcal{M}$ -subspaces of their coordinate functions.

Proposition 2.1. *Let $F = (f_1, \dots, f_m)$ be a vectorial Boolean function from $\mathbb{F}_2^n$ to $\mathbb{F}_2^m$. A subspace V of $\mathbb{F}_2^n$ is an $\mathcal{M}$ -subspace of F if and only if V is an $\mathcal{M}$ -subspace of each coordinate function f_i , $i = 1, \dots, m$.*

Proof. By definition, $D_a D_b F = (D_a D_b f_1, \dots, D_a D_b f_m)$, for all $a, b \in \mathbb{F}_2^n$. Hence, for all $x, a, b \in \mathbb{F}_2^n$, we have that $D_a D_b F(x) = 0_m$ if and only if $D_a D_b f_i(x) = 0$ for all $i = 1, \dots, m$. Consequently, V is an $\mathcal{M}$ -subspace of F if and only if it is an $\mathcal{M}$ -subspace of each coordinate function f_i . $\square$

Using this result, we can explicitly describe bent functions in the extended Maiorana-McFarland class of vectorial bent functions.

Theorem 2.2. *Let $n = 2k$ and $m \leq k$. The extended Maiorana-McFarland class $\mathcal{EM}$ coincides with the class of vectorial (n, m) -bent functions F of the form $F = (f_1, \dots, f_m)$ where*

$$f_i(x, y) = x \cdot \pi_i(y) + h_i(y), \quad (1)$$

for some $h_i \in \mathcal{B}_k$ and $\pi_i: \mathbb{F}_2^k \rightarrow \mathbb{F}_2^k$ such that $\langle \pi_1, \dots, \pi_m \rangle$ is an m -dimensional $\mathbb{F}_2$ -vector space of permutations. Consequently, F belongs to $\mathcal{EM}^\#$ if and only if F has at least one maximal $\mathcal{M}$ -subspace (shared by all coordinate functions).

Proof. Let $F(x, y) = \psi(x, y) + G(y)$ be an extended Maiorana-McFarland function. Then, by the definition, for each y there exists a linear map $L_y: \mathbb{F}_2^k \rightarrow \mathbb{F}_2^m$ such that $\psi(x, y) = L_y(x)$. Writing L_y coordinate-wise yields $\psi(x, y) = (x \cdot \pi_1(y), \dots, x \cdot \pi_m(y))$, for uniquely determined mappings $\pi_i: \mathbb{F}_2^k \rightarrow \mathbb{F}_2^k$. Hence $F = (f_1, \dots, f_m)$ with $f_i(x, y) = x \cdot \pi_i(y) + h_i(y)$,

where $G(y) = (h_1(y), \dots, h_m(y))$. A vectorial function is balanced if and only if all of its (nonzero) components are balanced (see, for example, Proposition 35 in [Car21]). For every fixed $x \in \mathbb{F}_2^k \setminus \{0_k\}$, the function $\psi(x, y) = (x \cdot \pi_1(y), \dots, x \cdot \pi_m(y))$ is balanced, therefore, it follows that every (nonzero) component of every (nonzero) linear combination of $\{\pi_1, \dots, \pi_m\}$ is balanced. Consequently, $\langle \pi_1, \dots, \pi_m \rangle$ is an m -dimensional $\mathbb{F}_2$ -vector space of permutations.

Conversely, let F be of the form $F = (f_1, \dots, f_m)$ where $f_i(x, y) = x \cdot \pi_i(y) + h_i(y)$, for some $h_i \in \mathcal{B}_k$ and $\langle \pi_1, \dots, \pi_m \rangle$ is an m -dimensional $\mathbb{F}_2$ -vector space of permutations on $\mathbb{F}_2^k$. Then, $x \mapsto \psi(x, y)$ is linear for every y , and for $x \neq 0_k$ and $v \neq 0_k$, the mapping

$$y \mapsto v \cdot \psi(x, y) = x \cdot \left(\sum_{i=1}^m v_i \pi_i(y) \right),$$

is balanced since $y \mapsto \sum_i v_i \pi_i(y)$ is a permutation. Therefore $F(x, y) = \psi(x, y) + G(y)$ belongs to the extended Maiorana-McFarland class.

The fact that F belongs to $\mathcal{EM}^\#$ iff F has at least one maximal $\mathcal{M}$ -subspace follows from Proposition 2.1 and the fact that the coordinate functions f_i given by Eq. (1) share the maximal $\mathcal{M}$ -subspace $\mathbb{F}_2^k \times \{0_k\}$. $\square$

Remark 2.3. *Computer simulations show that certain m -dimensional $\mathbb{F}_2$ -vector spaces of permutations are extendable to dimension $m + 1$, whereas others (of the same dimension) are not.*

In [LKM⁺25, Theorem 7] it was shown that the direct sum $f + g \in \mathcal{B}_{s+t}$ is outside $\mathcal{M}^\#$ if and only if at least one of the Boolean bent functions $f \in \mathcal{B}_s$ or $g \in \mathcal{B}_t$ is outside $\mathcal{M}^\#$. This statement can be easily extended to the vectorial case.

Corollary 2.4. *The direct sum $(x, y) \mapsto F(x) + G(y)$ of vectorial bent functions F and G is outside $\mathcal{EM}^\#$ if and only if at least one of the vectorial bent functions F or G is outside $\mathcal{EM}^\#$.*

Proof. This follows from Proposition 2.1 and [LKM⁺25, Theorem 7]. $\square$

3 Quadratic vectorial bent functions with and without common $\mathcal{M}$ -subspaces among all their coordinate functions

In this section, we show that in general quadratic vectorial $(n, 3)$ -bent functions may not share a maximal $\mathcal{M}$ -subspace. This is demonstrated by concatenating vectorial $(n, 3)$ -bent functions in a small number of variables from $\mathcal{GM}^\# \setminus \mathcal{EM}^\#$. To verify this exclusion property, we employ Algorithm 1, which extends the procedure for constructing $\mathcal{M}$ -subspaces of Boolean functions [PP20, Algorithm 1] to the vectorial case.

Algorithm 1 Construct the collection $\mathcal{MS}_r(f)$.

Require: A vectorial function $D_a D_b F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and $2 \leq r \leq n$.

Ensure: The collection $\mathcal{MS}_r(F)$.

- 1: **Construct** $\mathcal{MS}_2(F) := \{\langle a, b \rangle : \dim(U) = 2 \text{ and } D_a D_b F = 0_m\}$.
 - 2: **for all** subspaces $U \in \mathcal{MS}_2(F)$ **do**
 - 3: **repeat**
 - 4: **Determine** subspaces $\tilde{U} = \langle U, \tilde{u} \rangle$, for all $\tilde{u} \notin U$, such that for any two-dimensional vector subspace $\langle a, b \rangle \subseteq \tilde{U}$ the second-order derivative satisfies $D_a D_b F = 0_m$.
 - 5: **Put** $U \leftarrow \tilde{U}$ for the obtained subspaces $\tilde{U}$.
 - 6: **until** $\dim(U) = r$.
 - 7: **Output** subspaces U of dimension r .
 - 8: **end for**
-

Remark 3.1. *The only difference between Algorithm 1 and [PP20, Algorithm 1] is the initialization of the set of 2-dimensional $\mathcal{M}$ -subspaces, which is taken as the collection of common $\mathcal{M}$ -subspaces of the coordinate functions of F . The initialization used in Algorithm 1 above significantly reduces the number of admissible extensions at an early stage.*

Example 3.2. *Consider vectorial bent functions $F: \mathbb{F}_2^6 \rightarrow \mathbb{F}_2^3$, $G: \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^3$ and $H: \mathbb{F}_2^{10} \rightarrow \mathbb{F}_2^3$ with coordinate functions f_i, g_i and h_i , respectively, which are defined by their ANFs in the following way (we have abbreviated i for the variable x_i , and replaced 10 by 0 in h_1, h_2, h_3):*

- $f_1(x) = 14 + 25 + 36$, $f_2(x) = 15 + 16 + 24 + 25 + 34$, $f_3(x) = 13 + 14 + 26 + 45$,
- $g_1(x) = 15 + 26 + 37 + 48$, $g_2(x) = 13 + 14 + 34 + 25 + 45 + 36 + 46 + 17 + 37 + 47 + 28$, $g_3(x) = 12 + 13 + 14 + 24 + 16 + 26 + 56 + 17 + 27 + 37 + 18$,
- $h_1(x) = 10 + 12 + 02 + 03 + 14 + 04 + 24 + 15 + 35 + 45 + 06 + 36 + 46 + 56 + 17 + 27 + 37 + 57 + 67 + 08 + 28 + 38 + 48 + 68 + 78 + 19 + 09 + 29 + 49 + 59 + 69$,
- $h_2(x) = 10 + 02 + 13 + 23 + 14 + 04 + 15 + 35 + 16 + 26 + 36 + 17 + 07 + 27 + 67 + 18 + 08 + 28 + 48 + 78 + 19 + 29 + 49 + 69 + 79$,
- $h_3(x) = 13 + 03 + 23 + 14 + 04 + 24 + 34 + 15 + 25 + 35 + 45 + 16 + 06 + 26 + 36 + 46 + 27 + 47 + 57 + 67 + 68 + 09 + 39 + 49 + 69 + 79$.

The function G defined above is the third function in the list of EA-inequivalent quadratic vectorial $(8, 3)$ -bent functions obtained in [PP22]. The functions F and H (represented by their ANFs) in $n = 6$ and $n = 10$ variables can be viewed as 3-dimensional “bent subspaces” of vectorial bent functions of the form

$$X \in \mathbb{F}_{2^n} \mapsto (\text{Tr}_n(\beta_1 w X^3), \dots, \text{Tr}_n(\beta_{n/2} w X^3)) \in \mathbb{F}_2^{n/2},$$

where $(\beta_1, \dots, \beta_{n/2})$ is a basis of $\mathbb{F}_{2^{n/2}}$ over $\mathbb{F}_2$, and $w \in \mathbb{F}_{2^n}$ is not a cube. Such vectorial $(n, n/2)$ -bent functions were previously considered in [Car11, Section 3]. Using Algorithm 1, it is straightforward to verify that all these functions belong to $\mathcal{GM}^\# \setminus \mathcal{EM}^\#$, since none of them admits a maximal $\mathcal{M}$ -subspace.

Proposition 3.3. *For any $n \geq 6$, there exist quadratic vectorial $(n, 3)$ -bent functions in $\mathcal{GM}^\# \setminus \mathcal{EM}^\#$.*

Proof. The claim follows by applying Corollary 2.4 to the direct sum of the functions F , G , and H in Example 3.2. $\square$

Remark 3.4. *Using the known classifications for $n = 4, 6, 8$ (see [DMT19, PP21, PP22]), we verified via Algorithm 1 that every quadratic vectorial $(n, 2)$ -bent function admits a maximal $\mathcal{M}$ -subspace. Specifically, there is a unique EA-class for $n = 4$ and $n = 6$ admitting 5 and 9 such subspaces, respectively, while for $n = 8$ there are three EA-classes with 17, 21, and 85 maximal $\mathcal{M}$ -subspaces. This contrasts with the situation for quadratic vectorial $(n, 3)$ -bent functions, which do not, in general, possess a maximal $\mathcal{M}$ -subspace.*

4 On common $\mathcal{M}$ -subspaces of quadratic Boolean functions and their number

The results of the previous section indicate that, for small n , any two quadratic bent functions on $\mathbb{F}_2^n$ share at least $2^{n/2} + 1$ common $\mathcal{M}$ -subspaces of dimension $n/2$. In this section, we show that this property holds for all quadratic Boolean functions on $\mathbb{F}_2^n$ (not necessarily bent) for every even n . We note that the existence of a common $\mathcal{M}$ -subspace of dimension $n/2$ can be seen as a particular case of a more general, though not widely known, result of Alperin on common vanishing subspaces of two alternating bilinear forms [Alp65, Theorem 3]. For completeness and to keep the presentation self-contained using the terminology of (cryptographic) Boolean functions and $\mathcal{M}$ -subspaces, we provide a direct proof adapted to the present setting. We begin with the following auxiliary statement.

Lemma 4.1. *Let g be a quadratic Boolean functions in n variables, n odd. Then, there is a vector $v \in \mathbb{F}_2^n$ such that $D_w D_v g = 0$, for all $w \in \mathbb{F}_2^n$.*

Proof. Every quadratic Boolean function in an odd number of variables is extended affine equivalent to a function of the form

$$f(x_1, \dots, x_n) = x_1 x_2 + x_3 x_4 + \dots x_{2r-1} x_{2r}, \quad (2)$$

for some $r \leq \frac{n-1}{2}$; see, for example, Theorem 10 in [Car21]. The property that there exists a vector $v \in \mathbb{F}_2^n$ such that $D_w D_v g = 0$, for all $w \in \mathbb{F}_2^n$,

is invariant under extended affine equivalence. Consequently, it is sufficient to establish the theorem for functions of the form given in Eq. (2). Hence, suppose that f is of the form given in Eq. (2). For any non-zero vector in $\mathbb{F}_2^n$ of the form $v = (0_{2r}, v_{2r+1}, \dots, v_n)$, we have $D_w D_v f(x) = D_w(0) = 0$, for all $x, w \in \mathbb{F}_2^n$. This completes the proof. $\square$

With this result, we give a short proof of the existence of a common maximal $\mathcal{M}$ -subspace.

Theorem 4.2. *Let f and g be two quadratic Boolean functions in n variables, $n \geq 2$. There exists a common $\lfloor \frac{n+1}{2} \rfloor$ -dimensional $\mathcal{M}$ -subspace of f and g .*

Proof. We will prove the result by induction on n . The result is trivially true for $n = 2$. We assume now that the result is true for all k , $2 \leq k < n$ and prove that the result holds for n as well.

Case 1: n is even. Let V be any $(n-1)$ -dimensional subspace of $\mathbb{F}_2^n$. The functions $f' = f|_V$ and $g' = g|_V$ are Boolean functions in $(n-1)$ -variables, hence by the inductive hypothesis, there is a common $\mathcal{M}$ -subspace $W \subset V$ of f' and g' with

$$\dim(W) = \left\lfloor \frac{(n-1)+1}{2} \right\rfloor = \frac{n}{2} = \left\lfloor \frac{n+1}{2} \right\rfloor$$

(because n is even). For all $w_1, w_2 \in W$ and $x \in \mathbb{F}_2^n$ we have

$$\begin{aligned} D_{w_1} D_{w_2} f(x) &= D_{w_1} D_{w_2} f(0_n) = D_{w_1} D_{w_2} f'(0_n) = 0, \text{ and} \\ D_{w_1} D_{w_2} g(x) &= D_{w_1} D_{w_2} g(0_n) = D_{w_1} D_{w_2} g'(0_n) = 0. \end{aligned}$$

It follows that W is also a common $\mathcal{M}$ -subspace of f and g and we conclude that the result is true for n as well.

Case 2: n is odd. For odd n , from Lemma 4.1 it follows that there is a vector $v \in \mathbb{F}_2^n$ such that $D_w D_v f = 0$, for all $w \in \mathbb{F}_2^n$. The function $D_v g$ is affine, say $D_v g(x) = a \cdot x + b$, for some $a \in \mathbb{F}_2^n$ and $b \in \mathbb{F}_2$. Let $W_1 = \langle a \rangle^\perp$. For any $w \in W_1$, we have

$$D_w D_v g(0_n) = D_v g(w) + D_v g(0_n) = a \cdot w = 0.$$

Consequently, because g is quadratic, $D_w D_v g = 0$, for all $w \in W_1$.

In case $n = 3$, let a be a vector in $W_1 \setminus \{0_n, v\}$ (at least two such vectors exist since $\dim(W_1) = 2$). Then $\langle a, v \rangle$ is a common $\mathcal{M}$ -subspace of f and g and the result follows.

For $n > 3$, let W be any $(n-2)$ -dimensional subspace of W_1 not containing v . The functions $f' = f|_W$ and $g' = g|_W$ are Boolean functions in $(n-2)$ -variables, hence by the inductive hypothesis, there is a common $\mathcal{M}$ -subspace $Z \subset W$ of f' and g' with $\dim(Z) = \lfloor \frac{(n-2)+1}{2} \rfloor = \frac{n-1}{2}$. By the

construction, the subspace $V = \langle Z, v \rangle$ is a common $\mathcal{M}$ -subspace of f and g with

$$\dim(V) = \frac{n-1}{2} + 1 = \frac{n+1}{2} = \left\lfloor \frac{n+1}{2} \right\rfloor,$$

and we conclude that the result is true for n as well. $\square$

With this result, we are ready to explicitly determine the canonical form of quadratic vectorial $(n, 2)$ -bent functions.

Corollary 4.3. *Every quadratic vectorial $(n, 2)$ -bent function F has the following form, up to EA-equivalence:*

$$F(x, y) = \begin{pmatrix} x \cdot y + g(y) \\ x \cdot L(y) + h(y) \end{pmatrix}^T, \quad \text{where } x, y \in \mathbb{F}_2^{\frac{n}{2}}, g, h \in \mathcal{B}_{\frac{n}{2}}, \quad (3)$$

with $\deg(g), \deg(h) \leq 2$, and L is a linear complete permutation on $\mathbb{F}_2^{\frac{n}{2}}$ (i.e., both L and $L + \text{id}$ are permutations).

Proof. Due to existence of a common maximal $\mathcal{M}$ -subspace for both coordinate functions, it follows from Theorem 2.2 that F is EA-equivalent to the function of the form

$$(x, y) \mapsto \begin{pmatrix} x \cdot L_1(y) + g_1(y) \\ x \cdot L_2(y) + g_2(y) \end{pmatrix}^T,$$

where $x, y \in \mathbb{F}_2^{\frac{n}{2}}$, $g_i \in \mathcal{B}_{\frac{n}{2}}$, and $L_1, L_2, L_1 + L_2$ are linear permutations on $\mathbb{F}_2^{\frac{n}{2}}$. Applying the transformation $(x, y) \mapsto (x, L_1^{-1}(y))$ gives the canonical form given by Eq. (3). $\square$

The fact that in the proof of Theorem 4.2 the restriction chosen in the inductive step is essentially arbitrary allows us to obtain the following refinement of the result for even n .

Theorem 4.4. *Let $n \geq 2$ be an even number, and let f and g be two quadratic Boolean functions in n variables. Then, f and g share at least $2^{\frac{n}{2}} + 1$ common $\frac{n}{2}$ -dimensional $\mathcal{M}$ -subspaces.*

Proof. For $n = 2$ the result is trivially true. We assume $n \geq 4$ in the rest of the proof. From Theorem 4.2 it follows that there exists a common $\frac{n}{2}$ -dimensional $\mathcal{M}$ -subspace of f and g . Denote one such subspace by M_1 and set $\mathcal{A}_1 = \{M_1\}$. Assume that $1 \leq t \leq 2^{\frac{n}{2}}$ and that

$$\mathcal{A}_t = \{M_1, M_2, \dots, M_t\}$$

is a family of t distinct common $\frac{n}{2}$ -dimensional $\mathcal{M}$ -subspaces of f and g . We will show that there exists at least one common $\frac{n}{2}$ -dimensional $\mathcal{M}$ -subspace of f and g which is not contained in $\mathcal{A}_t$.

First, we will show that there exists an $(n - 1)$ -dimensional subspace of $\mathbb{F}_2^n$ (i.e., a hyperplane) which does not contain (as a subset) any of the subspaces from the family $\mathcal{A}_t$. Note that the total number of hyperplanes in $\mathbb{F}_2^n$ is $2^n - 1$. For a k -dimensional subspace $V \subset \mathbb{F}_2^n$, $0 \leq k \leq n$, the number of hyperplanes containing V is $2^{n-k} - 1$. Indeed, since this number is invariant under linear isomorphisms, we can assume that $V = \mathbb{F}_2^k \times \{0_{n-k}\}$. For a vector $a \in \mathbb{F}_2^n \setminus \{0_n\}$, the hyperplane defined by $a \cdot x = 0$ contains $\mathbb{F}_2^k \times \{0_{n-k}\}$ if and only if $a \in \{0_k\} \times \mathbb{F}_2^{n-k}$. Consequently, because the number of nonzero elements in $\{0_k\} \times \mathbb{F}_2^{n-k}$ is $2^{n-k} - 1$, we conclude that the number of hyperplanes containing V is $2^{n-k} - 1$. For $k = \frac{n}{2}$, we obtain that the number of hyperplanes containing an $\frac{n}{2}$ -dimensional subspace is $2^{\frac{n}{2}} - 1$. Consequently, the number of hyperplanes containing at least one subspace from $\mathcal{A}_t$ is at most $t(2^{\frac{n}{2}} - 1)$, and because $t \leq 2^{\frac{n}{2}}$, we have

$$t(2^{\frac{n}{2}} - 1) < (2^{\frac{n}{2}} + 1)(2^{\frac{n}{2}} - 1) = 2^n - 1.$$

Since $2^n - 1$ is the total number of hyperplanes in $\mathbb{F}_2^n$, it follows that there exists a hyperplane $W \subset \mathbb{F}_2^n$ that contains none of the subspaces in the family $\mathcal{A}_t$.

Applying Theorem 4.2 to the restrictions $f|_W$ and $g|_W$, it follows that they admit at least one common $\lfloor \frac{(n-1)+1}{2} \rfloor = \frac{n}{2}$ dimensional $\mathcal{M}$ -subspace. Denote one such subspace by M_{t+1} . In the same way as in the proof of Theorem 4.2, M_{t+1} is also an $\mathcal{M}$ -subspace of f and g . Note that M_{t+1} is contained in W and that W was chosen so that it contains none of the elements from $\mathcal{A}_t$, hence M_{t+1} is an $\frac{n}{2}$ -dimensional $\mathcal{M}$ -subspace of f and g which is not contained in $\mathcal{A}_t$. This completes the proof. $\square$

5 Conclusion

In this paper, we studied common $\mathcal{M}$ -subspaces for pairs of quadratic bent functions and determined the canonical form of quadratic vectorial $(n, 2)$ -bent functions. We also showed that there exist quadratic vectorial bent functions in the class $\mathcal{GM}^\#$ that do not belong to $\mathcal{EM}^\#$. These results raise several open questions that are important for a better understanding of vectorial bent functions. One natural direction is to construct quadratic vectorial bent functions in the class $\mathcal{EM}$; although such functions may be less interesting from a cryptographic point of view due to the presence of maximal common $\mathcal{M}$ -subspaces, they still give rise to non-trivial vector spaces of permutations, which can be used to construct vectorial bent-negabent functions as described in [AKM⁺25, PKPP23]. Finally, while some recent works (see, e.g., [PZKW21]) study constructions of vectorial bent functions whose component functions lie outside $\mathcal{M}^\#$ (and hence outside $\mathcal{GM}^\#$), to the best of our knowledge no constructions have been considered so far with respect to their relation to the classes $\mathcal{SM}$ and $\mathcal{EM}$.

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